FIELD PROJECT REPORT

(Semester III – B.Sc. Data Science)



University of Mumbai

PROJECT IDENTIFICATION

ANALYSIS OF
SEASONAL DISEASE PATTERNS
USING GOVERNMENT HEALTH DATA

A study of NCDC / IDSP outbreak-surveillance records in Maharashtra

B.Sc. DATA SCIENCE | SEMESTER III | FIELD PROJECT

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SEMESTER	**III**
INSTITUTION	**R.P. Institute of Hospitality & Management**
UNIVERSITY	**University of Mumbai**
ACADEMIC YEAR	**2026-2027**
PROJECT GUIDE	**Prof. Angelin**

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CERTIFICATE

This is to certify that **Mr. Khan Umar Abdulaziz,**
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R. P. Institute of Hospitality & Management, Vasai,
affiliated to the University of Mumbai, has successfully
completed the field project entitled

"Analysis of Seasonal Disease Patterns Using Government Health Data"

under the guidance of **Prof. Angelin,** during the academic year
2026 – 2027. This project is submitted in partial fulfillment of the
requirements for the Semester III examination of the
B.Sc. (Data Science) programme.

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College Stamp
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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me during the completion of this field project.

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I also extend my sincere thanks to **Prof. Khushali Sharma**, Head of the Department, for her constant motivation, valuable advice and support.

I am grateful to R. P. Institute of Hospitality & Management, Vasai, for providing the necessary resources and a conducive environment to undertake this project. I also acknowledge the University of Mumbai for the curriculum and the opportunity to work on this field project.

Lastly, I would like to thank my classmates, friends and family for their direct and indirect support during the course of this project.

DECLARATION

I, Khan Umar Abdulaziz, a student of S.Y. B.Sc. Data Science, Semester III, Roll No. 41, at R.P. Institute of Hospitality and Management, declare that I have prepared this field project report titled “Analysis of Seasonal Disease Patterns Using Government Health Data” for academic submission during the academic year 2026–2027.

I conducted the quantitative component using the NCDC/IDSP weekly outbreak-surveillance records and the analytical files maintained in my project repository. I have described the observation unit, data processing, statistical methods, findings and limitations so that the reader can distinguish government-reported outbreak records from individual patient counts and population incidence.

For the field component, I document the site context, key-informant questions and answers recorded in my field notes, and the observational details included in the project material. I have avoided identifying the informant by name and have treated photographs of public-health settings with care for privacy.

I have acknowledged the government data sources, repository outputs and background literature used in this report. The report is intended for academic evaluation and does not constitute clinical advice or a claim of causal proof about weather and disease transmission.

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ABSTRACT

Seasonal disease surveillance can help public-health teams prepare staff, diagnostics, vector-control activity and water-safety measures before reported outbreaks intensify. In this project, I analysed weekly outbreak-surveillance reports attributed to the National Centre for Disease Control (NCDC) and the Integrated Disease Surveillance Programme (IDSP) for Maharashtra. The main baseline covers the four completed calendar years 2022–2025 and contains 539 outbreak records across five selected disease families: dengue, acute diarrheal disease (ADD), malaria, food poisoning and chikungunya. The family-level tables sum to 20,955 reported cases and 266 reported deaths. I treated 2026 epidemiological Weeks 1–32 as a separate, partial comparison rather than a complete year.

I reviewed the repository’s data extraction, validation and disease-label normalization workflow, then compared monthly event distributions, seasonal concentration, year-to-year stability, district distribution and sensitivity to influential records. Dengue reporting was

concentrated from June to October, with October the largest monthly event count. ADD showed a June–August elevation, while malaria had a concentrated May–July window and was geographically clustered in eastern Vidarbha. Food-poisoning case totals were strongly affected by three high-volume February records. Chikungunya records were relatively sparse and did not support a stable statewide seasonal pattern.

I also documented a field visit to a public-health facility in the Nalasopara/Vasai–Virar area, including a key-informant interview and site observations, and related those qualitative notes to the limits of weekly government reporting. The study describes notified outbreaks rather than all illness in the population; it does not estimate incidence or prove that weather alone caused the observed patterns.

Keywords: seasonal surveillance, Maharashtra, NCDC, IDSP, dengue, malaria, data science, outbreak reporting, field observation.

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CHAPTER 1

1. INTRODUCTION

1.1 Background and Public-Health Rationale

Seasonal disease surveillance is important because the timing of reported outbreaks affects when public-health teams prepare diagnostic supplies, medicines, vector-control activity, risk communication and safe-water measures. Maharashtra includes dense urban corridors, peri-urban settlements, coastal areas, agricultural districts and forested regions. A statewide total can therefore conceal meaningful differences in timing and geographic concentration.

The south-west monsoon usually changes local environmental conditions through rainfall, humidity, water accumulation, drainage pressure and household storage practices. These conditions can coincide with changes in mosquito-borne and gastrointestinal illness. However, this study does not assume that any single weather event caused a reported outbreak. It asks a narrower, answerable question: how are officially notified outbreak records distributed over months and years, and how reliable are the resulting patterns?

1.2 What the Surveillance Records Represent

The central observation unit in my analysis is one government-reported outbreak record. A record is not necessarily one patient, one laboratory-confirmed infection or a complete count of all symptomatic people in a locality. It reflects an event that entered the formal notification system. Mild illness managed at home, unreported private-clinic visits and cases that never trigger a formal outbreak investigation may not be represented. Consequently, the totals and

percentages in this report describe reporting patterns, not population incidence, prevalence or individual risk.

1.3 Aim and Research Questions

My aim was to identify and compare seasonal patterns in five selected disease families using the government-reported outbreak records for Maharashtra. I asked: (1) which months contain the highest number of reported outbreak events for each family; (2) how stable these month rankings are across 2022–2025; (3) whether unusually large outbreak records distort case-volume peaks; (4) how disease records are distributed geographically; and (5) how field observations at a local public-health facility help explain the difference between daily clinical activity and weekly outbreak reporting.

1.4 Scope and Significance

The quantitative baseline includes four complete calendar years, 2022–2025. I kept the first 32 epidemiological weeks of 2026 as a separate comparison because later-year reporting is incomplete. I also built the companion research website (https://fieldprojects.vercel.app) to present the same repository-backed findings through interactive charts; the website is a communication layer, not an independent data source.



Figure 1. Research and data-processing workflow documented in the project.

CHAPTER 2

2. LITERATURE REVIEW

2.1 Seasonal Disease Patterns and Environmental Context

Public-health literature describes how reported disease patterns may vary with weather, water infrastructure, vector ecology, food handling, healthcare access and reporting practice. These factors provide context for a seasonal study, but the timing of an outbreak-notification record alone cannot identify which mechanism produced it. I use the literature to frame my

questions and interpret the results cautiously rather than claim that the repository measures weather, vector abundance or individual clinical causes.

Bhatt et al. (2013) mapped the global distribution and burden of dengue and demonstrated the value of examining disease patterns geographically as well as temporally. Gubler (2002) discussed dengue as a public-health problem shaped by urbanisation and social conditions. Curriero et al. (2001) examined associations between extreme precipitation and waterborne-disease outbreaks in the United States. I use these works as background for plausible mechanisms, not as proof that those mechanisms caused the patterns in my Maharashtra records.

2.2 Surveillance and Administrative Reporting

The Integrated Disease Surveillance Programme (IDSP), under the National Centre for Disease Control (NCDC) and the Ministry of Health and Family Welfare, is a key source of routine outbreak information in India. Shukla, Nayak and Mishra (2016) analysed nationwide IDSP outbreak reports from 2010–2015 and calculated seasonal indices from monthly averages. Their work is a close methodological precedent, although its national totals and categories are not directly comparable with my Maharashtra-only cohort. Raj, Soorya and Anitha Kumari (2022) examined IDSP vector-borne reports in Kerala and also emphasised the importance of data quality in interpretation.

2.3 Statistical Approaches Relevant to This Project

Seasonal indices help identify months with above-average reporting. I also use the coefficient of variation (CV), Gini coefficient and three-month concentration ratio (CR3) to describe the spread and concentration of monthly event counts and case totals. The Kruskal–Wallis H-test examines whether the observed monthly distributions differ, a Monte Carlo permutation test compares month counts with a uniform distribution, and Kendall's coefficient of concordance (W) assesses agreement among annual month-rank profiles. These tests address different questions and are not interchangeable. Kendall's W is interpreted cautiously because only four complete annual profiles are available.

The studies reviewed above answer related but different questions. Global dengue mapping provides geographic context; national IDSP analyses use the same broad type of notification source but combine many states; Kerala studies focus on vector-borne disease; and the Gondia study uses laboratory-confirmed dengue cases from one district. I therefore avoid comparing their raw totals directly with my figures. Instead, I use them to justify a Maharashtra-level study that preserves weekly source provenance, compares five disease

families over the same four-year baseline and separates outbreak frequency from reported case volume.

2.4 Comparative Relevance of the Reviewed Studies

Table 1. Literature reviewed and relevance to the present study

Study	Data and focus	Relevance to my project
Bhatt et al. (2013)	Global distribution and burden of dengue.	Provides geographic and epidemiological context; its estimates are not compared numerically with the Maharashtra outbreak records.
Curriero et al. (2001)	Extreme precipitation and waterborne-disease outbreaks in the United States.	Shows why water/environmental conditions may matter, but my dataset has no linked weather series to test causation.
Shukla, Nayak and Mishra (2016)	National IDSP outbreak reports for 2010–2015 and seasonal indices.	Closest methodological precedent; my study narrows the scope to Maharashtra and adds concentration, stability and outlier checks.
Raj et al. (2022)	IDSP vector-borne disease patterns in Kerala.	Demonstrates the value of state-level seasonal analysis and cautions that routine surveillance quality matters.
Sakure et al. (2026)	Laboratory-confirmed dengue at a Gondia medical college, 2023–2025.	Provides Maharashtra district context, but the clinical-confirmed case unit differs from my statewide outbreak-event unit.

2.5 Research Gap

The practical gap addressed by this project is the effort required to transform scattered weekly PDFs into an auditable multi-year dataset that distinguishes recurring temporal structure from label variation, missing reports, district concentration and exceptional high-case events. A district-focused study of laboratory-confirmed dengue in Gondia by Sakure et al. (2026) provides useful local context, but a clinical case series differs from the statewide outbreak-notification unit used here. My contribution is a consistent comparison of five disease families within one reporting archive, with event counts distinguished from case volumes and high-volume records examined through sensitivity analysis.

CHAPTER 3

3. PROBLEM STATEMENT AND PROBLEMS FACED

3.1 Problem Statement

Weekly disease-outbreak bulletins are published as PDFs rather than as one consistently structured analysis table. Before analysis, the source material must be extracted, checked against its original report, and normalized. If this work is skipped, spelling variants may split the same disease into separate categories, incomplete time periods may be compared unfairly, and isolated high-case events may dominate the apparent seasonal profile. This project responds to that problem by documenting a controlled processing workflow and comparing event frequency with reported case volume.

3.2 Practical Problems Encountered

Table 2. Practical problems encountered and responses

Problem	How I handled it
PDF extraction and table layout	Weekly bulletins required document-level extraction and manual review where line breaks or table formatting could alter a field.
Disease-label variation	Different spellings and related labels needed harmonization without discarding their original values.
Missing source weeks	The 2023 archive lacks published Weeks 15, 51 and 52. I did not create artificial zero-event records for absent reports.
Partial-year comparison	The 2026 dataset ends at Week 32 and omits later weeks when several disease families historically remain active.
Event count versus case volume	One large food-poisoning cluster can affect case totals much more than event counts; both measures were retained.
Uneven geographic reporting	Differences between districts may reflect burden, access, definitions and surveillance intensity as well as underlying disease dynamics.

3.3 How I Addressed These Problems

I maintained separate raw and normalized disease labels, preserved record provenance, compared outputs across analytical phases, kept the 2022–2025 baseline separate from partial 2026, and performed sensitivity checks on influential records. I also interpreted statistical significance cautiously because only four completed annual profiles are available for inter-annual comparisons.

CHAPTER 4

4. RESEARCH OBJECTIVES AND INQUIRY SCOPE

4.1 Objectives

Table 3. Research objectives and outcomes

No.	Objective	Project outcome
1	Assemble and validate Maharashtra outbreak records from weekly NCDC/IDSP reports.	Completed in repository analysis
2	Normalize disease labels and select five primary families for comparative analysis.	Completed
3	Describe monthly and annual outbreak-reporting distributions and identify peak windows.	Completed
4	Compare concentration and year-to-year stability across the 2022–2025 baseline.	Completed
5	Test the sensitivity of case-volume peaks to influential outbreak records.	Completed
6	Compare partial 2026 Week 1–32 reporting with prior equivalent periods.	Completed with temporal boundary

No.	Objective	Project outcome
7	Document local field context through a key-informant interview and site observations.	Field notes and site photographs documented
8	Translate the analysis into practical, evidence-led recommendations.	Completed

4.2 Study Area and Unit of Analysis

The quantitative study covers Maharashtra, including records attributed to its 36 districts. The field visit was focused on a public-health facility in the Nalasopara/Vasai–Virar area of Palghar district. These geographic scopes are not interchangeable: the statewide dataset describes reported events across the state, while the field component provides a local view of facility operations and surrounding infrastructure.

Table 4. Study scope and observation unit

Dimension	Specification
Quantitative geography	Maharashtra state, with district-level comparison
Primary period	2022–2025 completed calendar years
Secondary comparison	2026 epidemiological Weeks 1–32
Observation unit	One government-reported outbreak record
Primary families	Dengue, ADD, malaria, food poisoning, chikungunya
Field setting	Public-health facility and surrounding setting in Nalasopara / Vasai–Virar, Palghar

4.3 Delimitations

I did not estimate disease incidence per 100,000 population, conduct clinical diagnosis, or model weather variables as causal predictors. I treated field observations as local context and did not use them to calculate statewide rates. The 2026 partial-year comparison is not presented as a full-year forecast.

CHAPTER 5

5. METHODOLOGY AND ANALYTICAL FRAMEWORK

5.1 Research Design

I used a quantitative, descriptive and analytical design for the government data, supported by qualitative context from a key-informant interaction and a structured site walk-through. The strands serve different purposes. The data analysis identifies statewide patterns in notified

events; the field component helps describe how a local facility sees illness and processes information. Local observations do not alter the statewide event counts.

5.2 Data Processing Stages

Table 5. Data-processing pipeline and outputs

Stage	Work undertaken	Output
1. Source review	Organized weekly NCDC/IDSP PDFs by year and epidemiological week.	Archived source reports
2. Extraction	Extracted Maharashtra outbreak rows and associated fields.	809 extracted records across all disease families
3. Provenance and validation	Checked extracted rows against source reports and retained validation metadata.	Row-level audit trail
4. Harmonization	Mapped inconsistent disease names to canonical labels and analytical families.	30 analytical disease families
5. Cohort isolation	Selected five families with sufficient baseline observations for seasonality analysis.	539 records for 2022–2025
6. Aggregation	Aggregated records by month, epidemiological week, year and district.	Seasonal and spatial summary tables
7. Analytical tests	Calculated concentration metrics, seasonal indices, stability tests and sensitivity analyses.	Phase 5B–5D outputs
8. Interpretation	Compared findings and documented limitations before writing recommendations.	Evidence-graded conclusions



Figure 2. End-to-end data pipeline: from weekly reports to evidence-led interpretation.

5.3 Statistical Measures

Table 6. Statistical measures and their purpose

Measure	Purpose in my analysis
Monthly event count	Counts report records assigned to each calendar month.
Seasonal index	Compares each month's event frequency with the average monthly level; values above 1 indicate above-average reporting.
Three-month concentration ratio (CR3)	Measures the share of events captured within the selected three-month window.
Coefficient of variation (CV)	Measures relative dispersion of monthly counts around their mean.
Gini coefficient	Measures inequality in the distribution of monthly event counts.

Measure	Purpose in my analysis
Kruskal–Wallis test	Tests for differences in distribution across month groups; the result is interpreted alongside assumptions and sample size.
Kendall's W	Measures agreement among the four annual month-rank profiles; the p-value is treated as exploratory.
Outlier sensitivity	Recalculates monthly case patterns after excluding a small number of influential event records, solely as a sensitivity check.

5.4 Field Instrument and Ethics

For the field component I used a semi-structured key-informant interview format and a structured observation approach. I kept the informant anonymous, did not reproduce identifiable patient records, and used photographs for public-facility context rather than individual identification. The field questions focused on seasonal timing, diagnostic workflow, outbreak response, foodborne versus diarrheal presentations, reporting and opportunities to improve information flow.

CHAPTER 6

6. DATA COLLECTION AND PROCESSING PIPELINE

6.1 Primary Government Source

The core source is the NCDC/IDSP weekly outbreak surveillance archive published under the Ministry of Health and Family Welfare, Government of India. The analysed archive spans epidemiological Week 1 of 2022 through Week 32 of 2026 and contains 237 published weekly bulletins. The published 2023 archive has no reports for Weeks 15, 51 and 52; I did not turn these missing publications into artificial zero-outbreak weeks.

6.2 Dataset Profile

Table 7. Dataset profile and period coverage

Metric	Value used in this report
Extracted records across all disease families	809 overall (733 in 2022–2025; 76 in 2026 W1–32)
Primary disease-family records, 2022–2025	539
Reported cases in five baseline family tables	20,955 (sum of five family rows)
Reported deaths in five baseline family tables	266 (sum of five family rows)
Primary disease families	5
Partial 2026 records in the five families, Weeks 1–32	45 primary-family records
2026 reported cases summed across the five family rows	2,126 (sum of five family rows)

Metric	Value used in this report
2023 weeks missing from the published archive	2023 Weeks 15, 51 and 52 absent from the published archive; no artificial zeroes imputed

6.3 Disease Taxonomy and Selection

The extraction contained many variations in disease names. I retained source terminology for traceability and standardized the analysis labels so that variants such as “Dengue” and “Dengu”, or “ADD” and expanded diarrheal-disease spellings, could be grouped consistently. The main five-family cohort was selected using the repository’s baseline event thresholds and relevance to seasonal comparison.

6.4 Data Quality Decisions

I did not fill missing report weeks with artificial zeroes. I retained high-volume records in the baseline and then examined their influence through sensitivity analysis rather than deleting them from the source data. I also separated partial 2026 from complete years to avoid interpreting the absence of late-season weeks as a decline in annual disease activity.

The 2022–2025 family totals used consistently throughout the report are 204 dengue records, 153 ADD records, 70 malaria records, 69 food-poisoning records and 43 chikungunya records. These sum to 539 records, 20,955 reported cases and 266 reported deaths. For the partial 2026 cohort, the five family-level rows sum to 45 records, 2,126 reported cases and zero reported deaths. I use these arithmetic totals consistently in the results and appendices.

CHAPTER 7

7. ANALYSIS AND RESULTS: FIVE DISEASE SEASONAL SIGNATURES

7.1 Baseline Composition

The baseline contains 539 outbreak records. Dengue contributes the largest number of records (204; 37.8%), followed by ADD (153; 28.4%). Malaria and food poisoning contribute 70 (13.0%) and 69 (12.8%) records respectively, while chikungunya contributes 43 (8.0%). Record share describes the distribution of notified events, not relative population risk.

Table 8. Baseline profile of the five disease families (2022–2025)

Disease family	Records	Reported cases	Reported deaths	Share of records
Dengue	204	3,144	93	37.8%
Acute diarrheal disease (ADD)	153	9,050	93	28.4%

Disease family	Records	Reported cases	Reported deaths	Share of records
Malaria	70	2,133	51	13.0%
Food poisoning	69	5,927	29	12.8%
Chikungunya	43	701	0	8.0%
TOTAL	539	20,955	266	100.0%

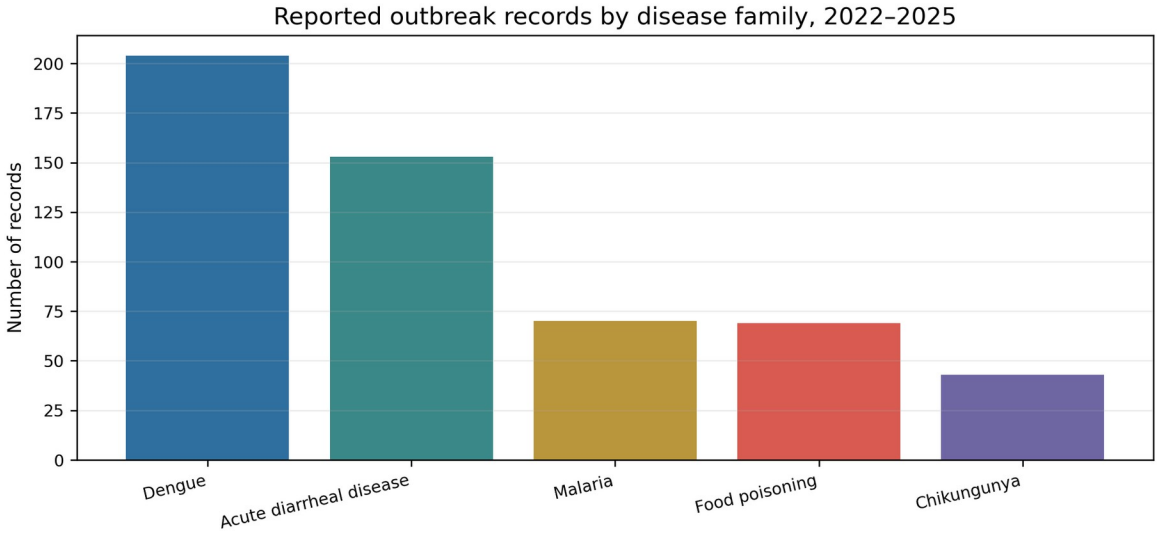


Figure 3. Baseline outbreak records by primary disease family.

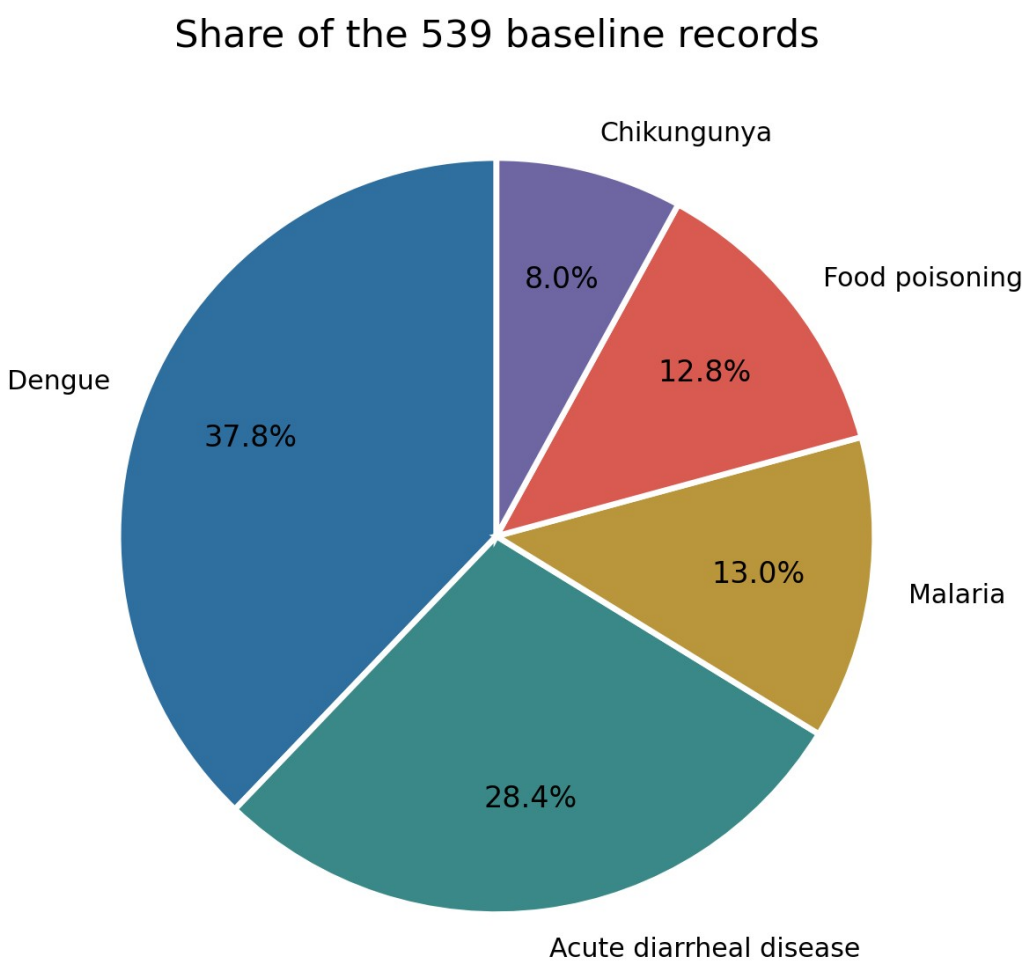


Figure 4. Share of the 539 baseline outbreak records by family.

7.2 Dengue: Strong Late-Year Concentration

Dengue is the largest baseline family, with 204 records, 3,144 reported cases and 93 reported deaths. The monthly distribution CSV contains 149 dengue records from June through October (149/204 = 73.0%). October has the largest pooled monthly count (40; 19.6%), followed by September (36; 17.6%). Under the equal-year-weighted seasonal-index method, however, September has the highest index (1.97), with October close behind (1.93). These summaries answer different questions: pooled counts give greater influence to high-volume years, while the seasonal index gives each completed year equal weight.

The repository reports monthly heterogeneity for dengue (Kruskal–Wallis H = 35.325, p = 0.00022); the Monte Carlo test of a uniform monthly distribution is also significant (p approximately 0.00010). Kendall's W is 0.405 (p = 0.086), so the annual month-rank profiles do not show statistically significant concordance under the reported approximation.

7.3 Acute Diarrheal Disease: Mid-Year Elevation

ADD is the largest family by reported case volume: 153 records, 9,050 cases and 93 reported deaths. The monthly distribution contains 57 records in June–August (19 in June, 22 in July and 16 in August), or 37.3% of the family total. The equal-year-weighted event seasonal indices are June 1.88, October 1.72, August 1.56 and July 1.47; October also has the largest pooled monthly count (25; 16.3%). The three largest pooled event months—October, July and June—sum to 66 records (43.1%), which is the top-three-month CR3 and should not be described as a contiguous June–August total.

Kendall's W is 0.530 (nominal p = 0.016; Benjamini–Hochberg q = 0.034), the strongest inter-annual rank concordance among the five families. This is exploratory rather than confirmatory evidence: the approximate p-value is based on only four annual profiles, and the result does not pass the stricter Bonferroni threshold of 0.00333 across the repository's 15 tests. The Kruskal–Wallis result for ADD is not significant (p = 0.132), while the Monte Carlo test against a uniform monthly distribution is significant (p approximately 0.00010); these tests have different null hypotheses.

7.4 Malaria: May–July Concentration with a Regional Footprint

Malaria contributes 70 records, 2,133 reported cases and 51 reported deaths. The May–July window contains 36 records (51.4%). June has the largest pooled monthly count (15; 21.4%), followed by July (14; 20.0%). The equal-year-weighted event seasonal indices are June 2.69, July 2.21 and May 1.59. The Kruskal–Wallis test gives H = 26.503 and p = 0.00546; this does not pass the repository's strict Bonferroni threshold of 0.00333, although its Benjamini–Hochberg q-value is 0.0142.

Geographically, Gadchiroli contributed 27 events and Chandrapur 16. Together they account for 43 of 70 malaria records (61.4%). This concentration matters: the aggregate Maharashtra curve is heavily influenced by eastern Vidarbha and should not be assumed to represent every district uniformly.

7.5 Food Poisoning: Bimodal Events and Outlier-Sensitive Cases

Food poisoning contributes 69 records, 5,927 reported cases and 29 reported deaths. The event distribution has two elevated windows: January–February accounts for 26.1% of records and April–May accounts for 29.0%. February and May tie at 11 pooled records each, while the equal-year-weighted event index is highest in May (2.51). By contrast, February contains 2,162 pooled reported cases, driven by a few high-volume records; the case peak should not be mistaken for a stable seasonal peak in event frequency.

The difference between event count and case volume is especially important. Three influential February records contribute 1,615 cases: one record in Kolhapur (651 cases, 2025 Week 6) and two records in Parbhani (629 and 335 cases, 2024 Week 6). Removing these records only for sensitivity analysis reduces February's case total from 2,162 to 547 and its share from 36.5% to 12.7%; April becomes the highest remaining case month with 899 cases. I retain the records in the main dataset and use the exclusion only to show how a few high-volume outbreaks affect the monthly case profile.

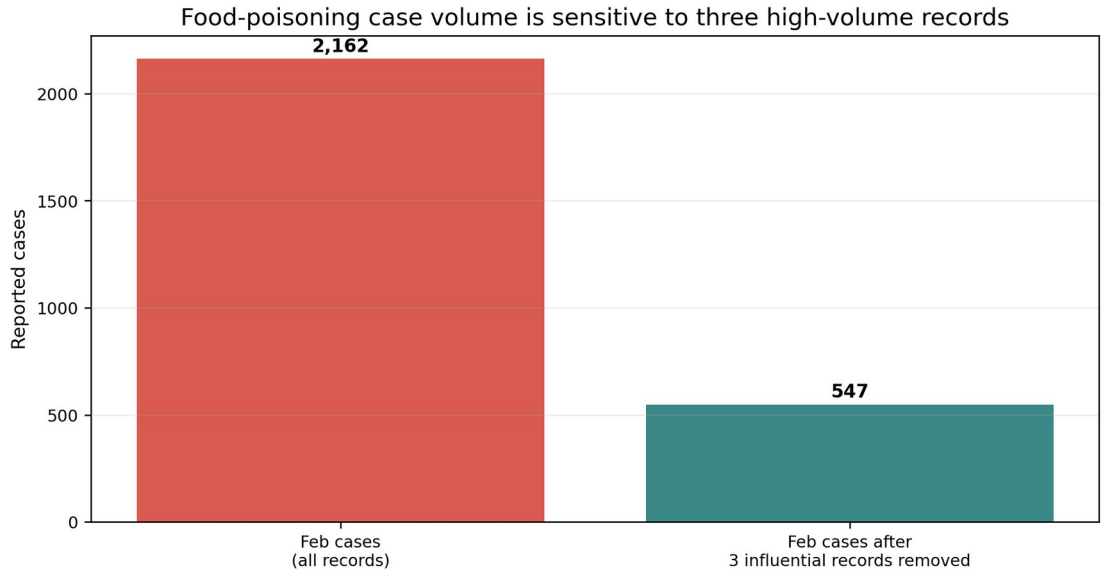


Figure 5. February food-poisoning case volume before and after the three-record sensitivity check.

7.6 Chikungunya: Sparse and Diffuse Evidence

Chikungunya has 43 baseline records, 701 reported cases and no reported deaths in the selected family table. May and June tie at six pooled events each; March, September and November have five each. The event CR3 is 39.5%, and Kendall's W is 0.150 ($p = 0.832$). Pune contributes 11 of 43 records (25.6%). The available records are sparse and do not establish a stable statewide seasonal pattern.

7.7 Comparative Seasonal Profile

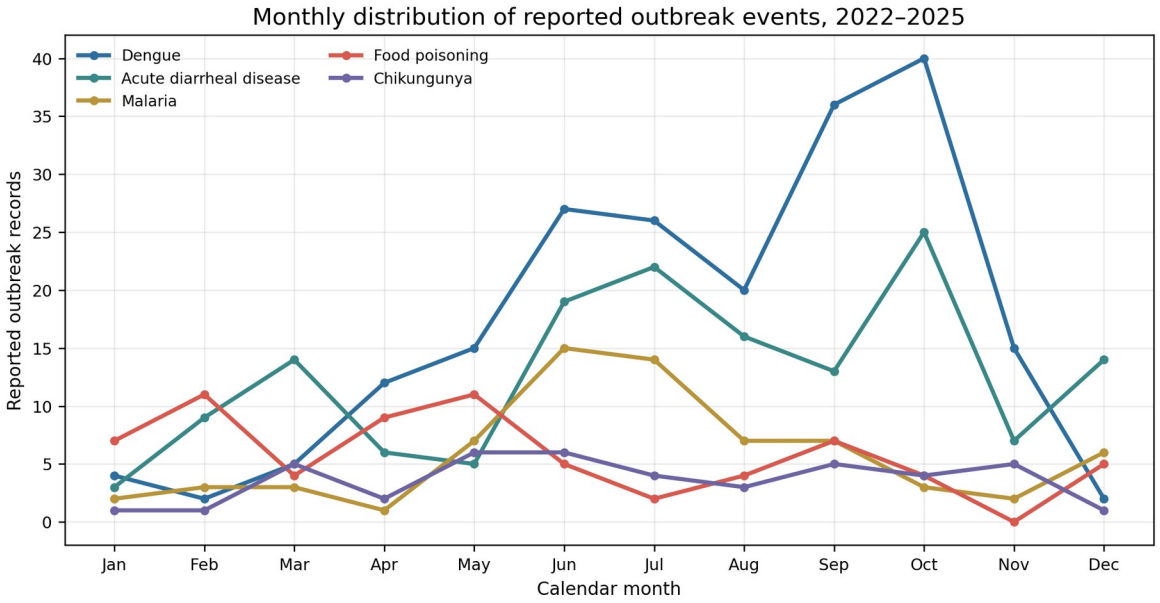


Figure 6. Monthly event counts for the five disease families, 2022–2025.

Table 9. Comparative seasonal signatures, concentration and stability

Family	Most informative timing pattern	CR3 / event-index CR3	Kendall's W (p)	Interpretation
Dengue	June–October = 149/204 (73.0%); pooled count peaks in October; equal-year index peaks in September	50.5% / 49.9%	0.405 (0.086)	Kruskal–Wallis p=.00022 and Monte Carlo p≈.00010; annual rank concordance is not significant.
ADD	June–August = 57/153 (37.3%); October has the highest pooled month count (25); top pooled months Oct, Jul, Jun = 43.1% CR3	43.1% / 47.1%	0.530 (0.016)	Kruskal–Wallis p=.132; Monte Carlo p≈.00010; W is exploratory (BH-FDR q=.034).
Malaria	May–July = 36/70 (51.4%); June is the pooled and equal-year-index peak	51.4% / 54.8%	0.319 (0.229)	Kruskal–Wallis p=.00546; Monte Carlo p≈.00010; reporting is concentrated in eastern Vidarbha.
Food poisoning	January–February and April–May; May has the highest equal-year-weighted event index	44.9% / 66.7%	0.386 (0.108)	Kruskal–Wallis p=.00568; Monte Carlo p=.0236; reported case totals are outlier-sensitive.
Chikungunya	May and June tie at six pooled events each; no stable contiguous window	39.5% / 52.5%	0.150 (0.832)	Kruskal–Wallis p=.3549 and Monte Carlo p=.4156; evidence is sparse and diffuse.

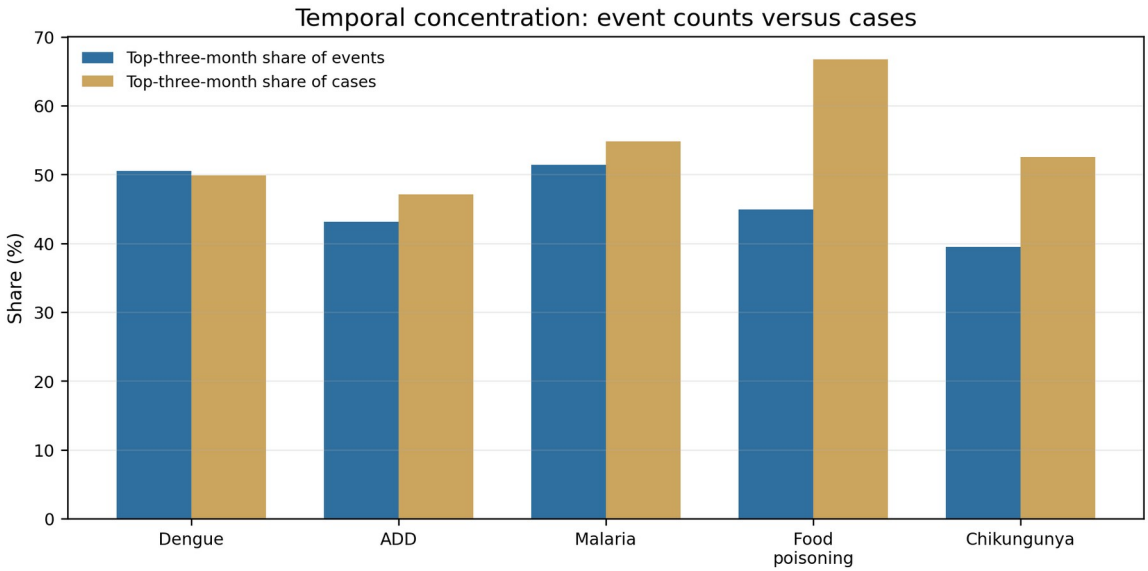


Figure 7. Three-month concentration for events compared with concentration for cases.

7.8 Annual Variation

The annual profiles show that the same number of years can contain very different event volumes. Dengue reported 20 events in 2022, 104 in 2023, 63 in 2024 and 17 in 2025. ADD rose to 71 records in 2024; food poisoning reached 29 in 2025; malaria stayed between 11 and 22; chikungunya's 22 records in 2024 represented more than half of its four-year baseline. These shifts are a reason to separate annual totals from a statement of predictable biological seasonality.

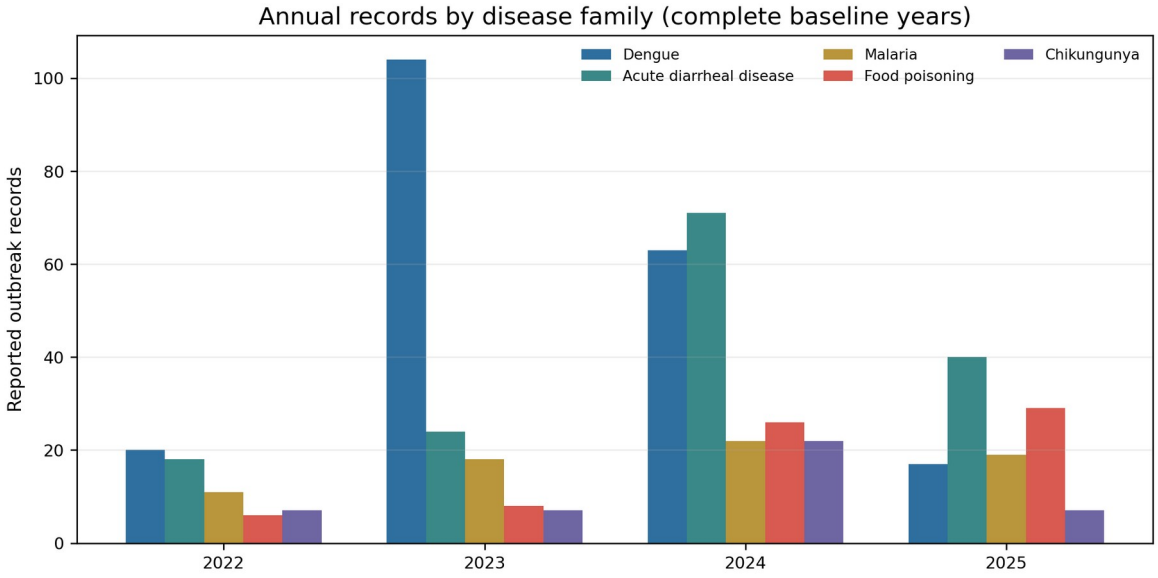


Figure 8. Annual reported outbreak records across the four complete baseline years.

7.9 Statistical Interpretation and Evidence Strength

The concentration statistics and monthly tests answer different questions. A high CR3 means the three busiest months contain a large share of records; it does not prove that the same peak

repeats every year. The Kruskal–Wallis H-test evaluates monthly heterogeneity, while the Monte Carlo test evaluates departure from a uniform monthly distribution. Their null hypotheses differ, so results need not match. The repository reports Kruskal–Wallis p-values of 0.00022 for dengue, 0.13207 for ADD, 0.00546 for malaria, 0.00568 for food poisoning and 0.35489 for chikungunya. The Monte Carlo test reports p-values of approximately 0.00010 for dengue, ADD and malaria, 0.02360 for food poisoning and 0.41556 for chikungunya. Kendall's W evaluates agreement among annual monthly ranks; because only four annual profiles are available, I treat the values as exploratory. ADD has the highest W (0.530; nominal p = 0.016; BH-FDR q = 0.034), but this does not pass the strict Bonferroni threshold of 0.00333 across the repository's 15 tests. I interpret each test alongside the event counts, year-to-year changes, geographic distribution and outlier sensitivity.

Table 10. Monthly dispersion and concentration metrics by disease family

Disease family	Event CV	Case CV	Event Gini	Case Gini	CR3: events / cases
Dengue	0.736	0.756	0.414	0.423	50.5% / 49.9%
ADD	0.524	0.609	0.298	0.344	43.1% / 47.1%
Malaria	0.753	0.802	0.395	0.450	51.4% / 54.8%
Food poisoning	0.564	1.160	0.315	0.534	44.9% / 66.7%
Chikungunya	0.515	0.783	0.289	0.421	39.5% / 52.5%

These measures complement one another: CV and Gini describe the unevenness of the monthly profile, while CR3 reports the share of total events or cases in the three highest months. Food poisoning has a notably higher case CR3 than event CR3, reflecting a small number of large reported clusters.

7.10 Partial 2026 Comparison through Week 32

I analysed the first 32 epidemiological weeks of 2026 separately from the completed-year baseline. The five family-level rows contain 45 records: six dengue, 18 ADD, one malaria, 19 food-poisoning and one chikungunya record. The corresponding case totals sum to 2,126. Compared with the historical mean for Weeks 1–32 of 2022–2025, the 2026 counts are six dengue versus 22.75; 18 ADD versus 21.75; one malaria versus 12.0; 19 food poisoning versus 12.75; and one chikungunya versus 6.25. This describes activity through Week 32 only. It does not include the later-year period, so lower dengue and malaria counts cannot be interpreted as full-year reductions.

Table 11. Partial 2026 family-level totals (Weeks 1–32)

Family	Records (2026 W1–32)	Reported cases	What the partial period shows
Dengue	6	106	Late-season period is incomplete; no full-year conclusion
ADD	18	864	Several records across the partial period; still not a full-year comparison

Family	Records (2026 W1–32)	Reported cases	What the partial period shows
Malaria	1	65	One reported event in the partial table; investigate locally, do not infer statewide absence
Food poisoning	19	1,082	Highest record count among five families in this partial window
Chikungunya	1	9	Too few records to infer a seasonal pattern
TOTAL	45	2,126	Partial-year comparison only

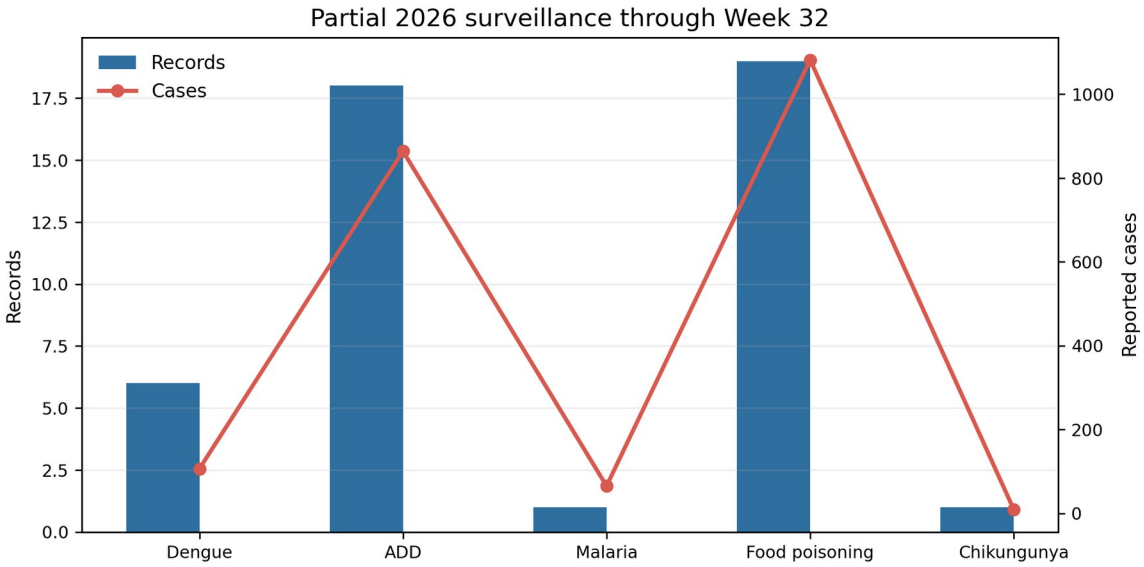


Figure 9. Partial 2026 family-level records and case totals through Week 32.

CHAPTER 8

8. FIELDWORK INVESTIGATION AND KEY-INFORMANT QUESTIONNAIRE

8.1 Field Visit and Facility Context

I visited the public-health facility in the Nalasopara West / Vasai–Virar area of Palghar district on 2 October 2026, with the observation period recorded from approximately 10:30 a.m. to noon. I wanted to understand how seasonal illness is encountered at facility level and how operational observations relate to the state surveillance data. The visible setting included a patient waiting corridor, reception/triage space and public-facing facility infrastructure. I treated the visit as local qualitative context, not as a representative survey of Maharashtra.

8.2 Structured Key-Informant Questionnaire: Questions and Answers

I used a semi-structured question set with the Medical Officer-in-Charge. The questions covered seasonal disease timing, diagnostic processes, reporting pathways, outbreak response, foodborne illness and possible improvements to data flow. I paraphrase the answers

in my field notes rather than present them as verbatim quotations, and I do not identify the informant by name.

Q1. Seasonal shifts: What seasonal shifts in patient volume and disease presentation do you observe throughout the year?

My notes record two broad periods described by the officer: gastrointestinal illness and diarrhoea increasing around the June–July onset of the rains, and fever presentations, including clinically suspected dengue and viral fever, becoming more prominent from September to November. The officer also distinguished the peri-urban setting of Nalasopara from the forested parts of Palghar district.

Q2. Vector ecology: How does local vector-borne transmission differ from rural or forested districts?

The officer distinguished dense peri-urban Nalasopara from forested districts elsewhere in Palghar. The response associated local malaria concerns with construction-related standing water and dengue concerns with domestic and peri-domestic containers, including storage drums and overhead cisterns.

Q3. Diagnostic workflow: How are suspected outbreak diseases diagnosed and recorded at the facility?

The officer described clinical assessment and rapid diagnostic tests, including malaria antigen cards and dengue NS1/IgM tests. Where indicated, samples are sent to a sub-district or district reference laboratory for additional confirmation.

Q4. Reporting hierarchy: How does information move from outpatient registration to IDSP reporting?

The officer described daily entries in physical outpatient registers, consolidation of fever and diarrhoea information, classification into presumptive and laboratory-confirmed reporting streams, and periodic submission to the Palghar District Surveillance Unit.

Q5. Containment triggers: What happens when several cases cluster in one neighbourhood?

The officer described notifying municipal sanitation teams when a local cluster is identified, checking water-storage practices, distributing chlorine tablets, conducting mosquito-control activity and arranging oral rehydration support.

Q6. Logistical hurdles: What difficulties arise during seasonal peaks?

The officer identified pressure on rapid-test supplies during sudden fever surges, delays in confirmatory laboratory results and delayed presentation to public facilities when people first self-medicate or consult private pharmacies.

Q7. Clinical presentation: How does food poisoning differ from general acute diarrhoeal disease?

The officer distinguished sudden groups of people becoming ill within hours after a shared meal or community event from diarrhoeal cases that appear across separate households over a longer period. This distinction is relevant to my analysis because one

high-volume food-poisoning event can change reported case totals without a proportionate increase in the number of outbreak records.

Q8. Summer heat: Does hot weather affect foodborne or gastrointestinal presentations?

The officer identified hotter months as a period when unsafe food handling, unrefrigerated dairy and stored drinking water may create additional food-safety concerns. I treat this as the informant's operational perspective: my dataset contains no linked temperature or food-handling measurements, so it cannot test this explanation causally.

Q9. Surveillance gaps: What is missed between daily clinical activity and official weekly reports?

The officer explained that official bulletins aggregate investigated outbreak clusters, while routine mild cases may remain in facility registers and never appear as separate outbreak notifications unless a cluster meets reporting criteria. This aligns with my observation unit: a notified outbreak record rather than a line list of every patient.

Q10. Technology: What change would improve facility-level surveillance?

The officer favoured direct digital entry at the triage point, automated alerts when counts rise and better integration between facility entries and reference-laboratory results. The intended benefit would be less duplicate transcription and a more timely route from a local signal to district-level review.

8.3 Structured Site Observations

I recorded three main observations during the facility walk-through. They describe the setting and workflow visible during one visit; they are not a statistical sample of patients or a formal facility audit across multiple dates.

Table 12. Field observations recorded during the visit

Time / zone	Observation recorded
10:40 AM — Outpatient corridor	High morning activity was visible in the waiting corridor, and patient triage appeared to rely on manual workflow. My field note estimated that a substantial portion of people waiting had fever or gastrointestinal complaints; this was a one-time visual estimate and not a measured clinical statistic.
11:15 AM — Point-of-care area	The field note described an accessible oral rehydration area and rapid diagnostic-test materials for malaria and dengue. The observation was limited to visible site context and did not constitute an independent stock audit.
12:00 PM — Facility perimeter and nearby housing	The field note described roadside drainage with intermittent stagnant water and domestic water storage in large plastic drums in the surrounding residential setting. The potential significance is as a context for environmental-health questions, not as proof that these features caused a specific reported outbreak.

8.4 What the Field Visit Added to the Data Analysis

The most useful contribution of the visit was contextual rather than statistical. The interview notes describe how daily clinical activity may be more detailed than the weekly outbreak

bulletin, because the bulletin captures notified clusters rather than all outpatient visits. The reporting pathway—from facility registers to district-level aggregation—also clarifies why a weekly publication date is not identical to the date an illness first appeared.

The field notes also help interpret the difference between ADD and food poisoning. A prolonged spread of gastrointestinal illness across households creates a different reporting pattern from a single shared exposure that produces a large cluster in a short period. This is consistent with the quantitative sensitivity analysis, in which a few large food-poisoning records change the monthly case-volume distribution substantially.

8.5 Link between Field Context and Statewide Surveillance

Table 13. Field context compared with statewide surveillance

Field-context theme	Relevant quantitative finding	How I interpret the connection
Monsoon-associated gastrointestinal presentations	ADD records rise in June–August.	Local context is compatible with the timing, but does not prove a cause.
Late-monsoon and post-monsoon fever concerns	Dengue records concentrate June–October, with October peak.	The interview perspective is directionally consistent with the statewide notification profile.
Shared-meal clusters	Food-poisoning case totals are highly sensitive to three records.	Point-source episodes can increase case totals without a proportional increase in event frequency.
Daily register and weekly notification pathway	The study uses weekly outbreak PDFs.	Operational reporting thresholds and aggregation mean clinical workload and bulletin counts are different measures.
Standing water and household storage	Malaria is concentrated in eastern Vidarbha; dengue is widely distributed.	Local environmental context is relevant to prevention planning but is not an explanatory variable measured in this dataset.

CHAPTER 9

9. CONCLUSION AND SUGGESTIONS

9.1 Conclusion

Through this project, I organised the NCDC/IDSP weekly outbreak reports into a structured dataset and compared five disease families across the complete 2022–2025 baseline. The cohort contains 539 records, 20,955 reported cases and 266 reported deaths. Dengue has the largest pooled monthly event count in October, while the equal-year-weighted event seasonal index peaks in September; 149 records (73.0%) occur from June through October. ADD has a June–August elevation (57 records, 37.3%) and a separate October increase; malaria is concentrated in May–July and strongly represented by Gadchiroli and Chandrapur; food-poisoning case totals are sensitive to a handful of high-volume events; and chikungunya records are too sparse to establish a stable statewide seasonal pattern.

The main methodological lesson is that the frequency of outbreak records and the number of reported cases are not interchangeable. The top-three-month share of food-poisoning cases is 66.7%, compared with 44.9% of its records. The sensitivity check shows why a few very large records can dominate a case-based curve. I therefore interpreted event concentration, case concentration, statistical results, district footprint, annual variation and sensitivity checks together. I also kept partial 2026 records separate from complete years so unobserved later weeks would not be mistaken for a decline in annual activity.

The field visit added a local operational perspective on patient flow, rapid testing, municipal response and weekly reporting. It supports the importance of linking administrative data with the realities of facility workflow, but it does not establish statewide prevalence or causal effects. The project's overall contribution is a transparent, reproducible description of government-reported outbreak seasonality and a set of practical questions for future surveillance improvement.

9.2 Recommendations

Table 14. Recommendations linked to the findings

Recommendation	Application
1. Prepare ahead of the early monsoon	Use the May–June period to review water-safety communication, ORS availability, outbreak triage and reporting readiness before the observed ADD elevation.
2. Sustain vector-control activity through late monsoon	Because dengue notifications are concentrated June–October, source reduction, household container management and public communication should extend beyond the first rains.
3. Use district-sensitive malaria planning	The high share of records from Gadchiroli and Chandrapur suggests that district-level planning is more informative than a uniform statewide interpretation.
4. Separate event alerts from case-volume alerts	A small number of mass-exposure food-poisoning events can affect case totals sharply; dashboards should display record counts and case counts side by side.
5. Improve data-entry and notification linkage	Explore digital support to reduce repeated transcription from facility records to district-level reporting, while preserving verification and audit trails.
6. Keep incomplete periods clearly bounded	Interim 2026 dashboards should show the last observed week prominently and avoid treating early-year counts as full-year outcomes.
7. Maintain consistent disease labels	Store raw source strings beside canonical family labels and review label shifts periodically to protect trend comparability.
8. Continue local field documentation	Future site visits can use anonymized questionnaires, observation checklists and consent-based notes to compare local perception with the surveillance calendar.

9.3 Limitations and Future Work

The records are outbreak notifications rather than individual case line lists and have no population denominator for incidence calculations. The 2023 source archive has three missing weekly bulletins, and the partial 2026 period excludes later weeks. Reporting

intensity, case classification and district practices may vary by year. Kendall's W is estimated from only four annual profiles; p-values are exploratory and do not establish a biological mechanism or a causal role for rainfall. The ADD table also contains an anomalous concentration of reported deaths in December, including repeated values that do not align with the corresponding small case counts; I therefore do not interpret those rows as a reliable seasonal fatality pattern. The field component is localized to one facility setting and is not representative of all districts.

Future work could extend the baseline as newer weekly reports become available, rerun the analysis from a versioned machine-readable extract, compare notification timing with independent weather observations using a prespecified lag structure, and expand fieldwork across more settlement types and districts. Completed community questionnaires and any additional interviews should be retained as anonymised records so qualitative summaries can be reviewed alongside the quantitative results.

CHAPTER 10

10. REFERENCES AND BIBLIOGRAPHY

10.1 Government and Data Sources

1. National Centre for Disease Control (NCDC), Directorate General of Health Services, Ministry of Health and Family Welfare, Government of India. Weekly Outbreak Surveillance Reports / Integrated Disease Surveillance Programme (IDSP), Maharashtra records, 2022–2026. https://ncdc.mohfw.gov.in/
2. Integrated Disease Surveillance Programme (IDSP). Programme information and surveillance resources. Government of India. https://idsp.nic.in/
3. Ministry of Health and Family Welfare / IDSP. IDSP outbreak master archive, hosted by Dataful (national dataset and metadata). https://dataful.in/datasets/18514
4. Project repository. Field Project: Analysis of Seasonal Disease Patterns Using Government Health Data. https://github.com/RizzyCoder19/Field-Project
5. Project website. Seasonal disease pattern analysis in Maharashtra. https://fieldprojects.vercel.app/
6. Project repository artefacts used: Phase 5B seasonal-analysis report and monthly/yearly distribution tables; Phase 5C advanced-analysis and outlier-sensitivity tables; Phase 5D interpretation, validation and limitation documents; fieldwork methodology, questionnaire, interview guide, observation checklist and ethics protocol.

10.2 Peer-Reviewed Literature

7. Bhatt, S., Gething, P. W., Brady, O. J., Messina, J. P., Farlow, A. W., Moyes, C. L., et al. (2013). The global distribution and burden of dengue. Nature, 496, 504–507.

8. Curriero, F. C., Patz, J. A., Rose, J. B., & Lele, S. (2001). The association between extreme precipitation and waterborne disease outbreaks in the United States, 1948–1994. American Journal of Public Health, 91(8), 1194–1199.

9. Gubler, D. J. (2002). Epidemic dengue/dengue hemorrhagic fever as a public health, social and economic problem in the 21st century. Trends in Microbiology, 10(2), 100–103.

10. Kendall, M. G., & Babington Smith, B. (1939). The problem of m rankings. The Annals of Mathematical Statistics, 10(3), 275–287.

14. Sakure, K., Tankhiwale, S., Patil, V., & Khadse, R. (2026). The demographic and seasonal variation of dengue virus infection in Gondia, a tribal district in Maharashtra, India. International Journal of Current Pharmaceutical Review and Research, 18(2), 1532–1537. https://doi.org/10.25258/ijcpr.18.2.253

13. Raj, J., Soorya, V., & Anitha Kumari, K. R. (2022). Trends and seasonality of vector-borne diseases in Kerala. International Journal of Community Medicine and Public Health, 9(10), 3701–3709. https://doi.org/10.18203/2394-6040.ijcmph20222559

12. Shukla, D., Nayak, P., & Mishra, R. N. (2016). Burden of infectious diseases and their seasonal variation in India. International Journal of Health Sciences and Research, 6(12), 33–40.

11. Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. Journal of the American Statistical Association, 47(260), 583–621.

CHAPTER 11

11. PROJECT PLANNER / EXECUTION TIMELINE

I organized the project as a sequence of source review, data engineering, statistical analysis, interpretation, field contextualization and report preparation. The following planner records the main work stages and their outputs.

Table 15. Project planner and major outputs

Stage	Work completed	Output
Weeks 1–2	Review source archive and repository; organize weekly PDFs and verify source fields.	Source register and extraction plan
Weeks 3–4	Extract Maharashtra rows, check provenance and normalize disease labels.	Validated master records and label map

Stage	Work completed	Output
Weeks 5–6	Build the baseline, monthly/annual profiles, concentration measures and stability tests.	Phase 5B and Phase 5C outputs
Week 7	Run outlier sensitivity checks, district comparisons and partial-2026 analysis.	Sensitivity tables and validation summary
Week 8	Complete field-context notes, recommendations, visualizations and report.	Report, figures, references and appendices

11.1 Project Deliverables

The final project package includes the research report, source and analytical files in the GitHub repository, the interactive project website, statistical charts, questionnaire and interview instruments, field-site photographs, and this written account of the methods and findings.

CHAPTER 12

12. GUIDE INTERACTION DIARY / SUPERVISION RECORD

I organised this review record around the main stages of the project and the methodological decisions reflected in the report. It records the workstream and the action taken rather than assigning dates or signatures to undocumented meetings.

Table 16. Project review topics and actions

Project review stage	Review focus	Action / learning recorded
Stage 1: scope and sources	Research question and NCDC/IDSP source boundary	I defined Maharashtra as the study area, made the outbreak record the unit of analysis, and retained the original source files for traceability.
Stage 2: data processing	Extraction, validation and disease-label harmonisation	I kept original source strings beside standardized labels and used the row-level audit trail to preserve provenance.
Stage 3: seasonal analysis	Seasonal index, concentration measures and statistical tests	I distinguished pooled monthly counts from equal-year-weighted indices and interpreted significance alongside the small number of annual profiles.
Stage 4: sensitivity analysis	Influential food-poisoning, ADD and malaria records	I retained the source records and used exclusion only as a sensitivity test to show the effect of influential case volumes.
Stage 5: field context and report	Interview notes, site observations, visuals and recommendations	I kept local contextual material separate from statewide numerical estimates and used privacy-conscious treatment of the waiting-area image.

CHAPTER 13

13. APPENDICES

Appendix A. Rapid Field Questionnaire

The following concise questionnaire is designed for a voluntary, anonymous conversation with a community member or local stakeholder. It can be answered through selected response options and does not request names, phone numbers, patient details or other identifying information.

1. In your experience, do acute illnesses in your locality increase during particular months?
2. Which period seems to have the highest illness activity: summer, early monsoon, late monsoon, post-monsoon or winter?
3. Which of the following have you observed or heard diagnosed locally: dengue, malaria, ADD, food poisoning, chikungunya?
4. In which months do mosquito-borne fevers seem most visible in your area?
5. When do diarrhoea or stomach infections seem to increase?
6. Have you noticed a localized group of people becoming ill after a common meal or gathering?
7. What environmental condition is most visible near the residence or workplace: stagnant drains, water-filled potholes, construction water, uncovered storage, or none seen?
8. What is the main drinking-water source: municipal tap, tank, tanker/standpost, packaged/filtered water or well/borewell?
9. How often do you notice mosquito-control or sanitation activity in the locality?
10. Have you seen public-health messages about preventing dengue, malaria or diarrhoeal disease?
11. If someone develops fever, where do they usually seek help first: public facility, private clinic, pharmacy or home care?
12. What usually encourages people to seek medical attention?
13. Are diagnostic tests easy to access locally?
14. Are public-health workers or community workers visible in the locality?
15. How quickly do residents hear about a local outbreak or prevention drive?
16. What is the most difficult part of getting care during an illness surge?
17. Have you noticed waterlogging or drainage problems after heavy rain?
18. What could the municipality improve before the monsoon?

19. Would simple early-warning messages by SMS, WhatsApp, posters or local announcements be useful?

20. Is there anything else about seasonal illness or local response you would like to share?

Appendix B. Key-Informant Interview Record

This appendix lists the interview domains used during the field visit. The full question-and-answer record is presented in Chapter 8 to keep the main narrative close to the field context.

Table 17. Key-informant interview domains

Domain	Questions covered
Seasonal chronology	Seasonal shifts; differences between dengue and malaria timing; gastrointestinal illness timing.
Diagnosis and reporting	Rapid tests and laboratory confirmation; facility registers; reporting to the district surveillance unit.
Outbreak response	Local cluster recognition; coordination with sanitation teams; chlorine/ORS/vector-control response.
Operational constraints	Rapid-test supply; confirmatory laboratory turnaround; care-seeking before reaching public facilities.
Foodborne illness	Shared-meal clusters versus cases distributed over time; food safety during hot months.
Data-system improvement	Digital entry at triage, automated thresholds and integration with laboratory results.

Appendix C. Observation Checklist and Summary

I structured the site walk-through around four practical areas: outpatient flow, visible testing/support areas, facility access and the immediate surroundings. The summary below records the observations documented for the visit.

Table 18. Field-observation checklist summary

Area	Indicator	Summary recorded
OPD corridor	Patient flow and triage	High morning activity; manual workflow visible.
Support area	ORS and rapid-test materials	ORS corner and rapid-test materials described in field notes.
Facility boundary	Drainage and standing water	Intermittent stagnant water noted in nearby drainage.
Residential surroundings	Water storage containers	Large plastic drums visible/described as household storage in the surrounding setting.
Communication	Public-health information	Public-facing setting was documented photographically; a systematic IEC poster count was not part of the recorded field note.

Appendix D. Field-Site Photographic Documentation

I use the photographs below to illustrate the public-facing facility and surrounding setting. They document site context rather than a measured environmental audit. Where a timestamp

is visible, I retain it as the image date and do not assume it matches the October 2026 visit. Faces in the waiting-area image have been softened for privacy.



Figure 10. Site-context photo panel: municipal health-facility exterior, covered entrance, outpatient corridor (faces softened) and surrounding roadside setting.

Table 19. Description of site-context photographs

Panel	What is shown
D1	Public-health facility exterior and signage.
D2	Covered public-facing entrance and approach.
D3	Indoor corridor/service counter; the image shows a timestamp of 22 March 2022, so it documents context and is not treated as a photograph of the 2 October 2026 visit.
D4	Roadside and drainage context shown in the supplied location image; it is illustrative and is not treated as a measured environmental audit.

Appendix E. Summary Data Tables

The table below consolidates the principal numerical values used in the findings. Percentages refer to the event records or reported case values in the source analysis, not population incidence.

Table 20. Summary of findings by disease family

Family	Records	Cases	Deaths	Seasonal feature
Dengue	204	3,144	93	149/204 (73.0%) in Jun–Oct; pooled event peak Oct; equal-year-weighted event index peak Sep

Family	Records	Cases	Deaths	Seasonal feature
ADD	153	9,050	93	57/153 (37.3%) in Jun–Aug; October pooled count peak; June equal-year event index peak; top-three-month CR3=43.1%
Malaria	70	2,133	51	36/70 (51.4%) in May–Jul; 43/70 (61.4%) in Gadchiroli and Chandrapur
Food poisoning	69	5,927	29	Jan–Feb and Apr–May event windows; three records drive much of February case volume
Chikungunya	43	701	0	Sparse and diffuse; W=0.150, p=.832; no stable statewide pattern established

Appendix F. Viva Preparation: Concise Responses

Table 21. Viva preparation questions and responses

Question	Response
What is the project about?	I studied seasonal patterns in government-reported disease-outbreak records in Maharashtra using data processing and statistical analysis.
Why use NCDC/IDSP records?	They provide weekly reports with geography, disease labels and outbreak counts that can be compared over time.
What is the most important unit distinction?	One outbreak record is not one patient and does not measure population incidence.
Which disease showed the strongest late-year concentration?	Dengue has 149 of 204 records (73.0%) from June through October. October has the largest pooled count (40); the equal-year-weighted seasonal index peaks in September.
Why did I analyse food poisoning outliers?	Three high-volume records contributed 1,615 cases and materially changed February case totals.
Why is 2026 not compared as a full year?	The analysis stops at Week 32, before much of the late-monsoon/post-monsoon period.
What did the field visit add?	It provided context about facility workflow, diagnostic support, environmental setting and the difference between daily clinical activity and weekly outbreak notification.
What is one major limitation?	The source captures notified outbreak clusters, lacks population denominators and cannot establish causal effects from seasonality alone.
Why were the five disease families selected?	The repository audit selected families with sufficient baseline event counts and relevance to seasonal comparison.
Why keep raw disease labels?	Keeping source strings preserves traceability and lets later reviewers check whether normalization decisions were reasonable.
Why use an outlier sensitivity check?	It demonstrates whether the headline monthly case peak depends heavily on a few very large records.
What are my main recommendations?	Use disease-specific timing rather than one single “disease season,” show event counts separately from case volumes, extend the baseline as reports accumulate, and collect documented field evidence across more than one locality.